TEST &
VALIDATION



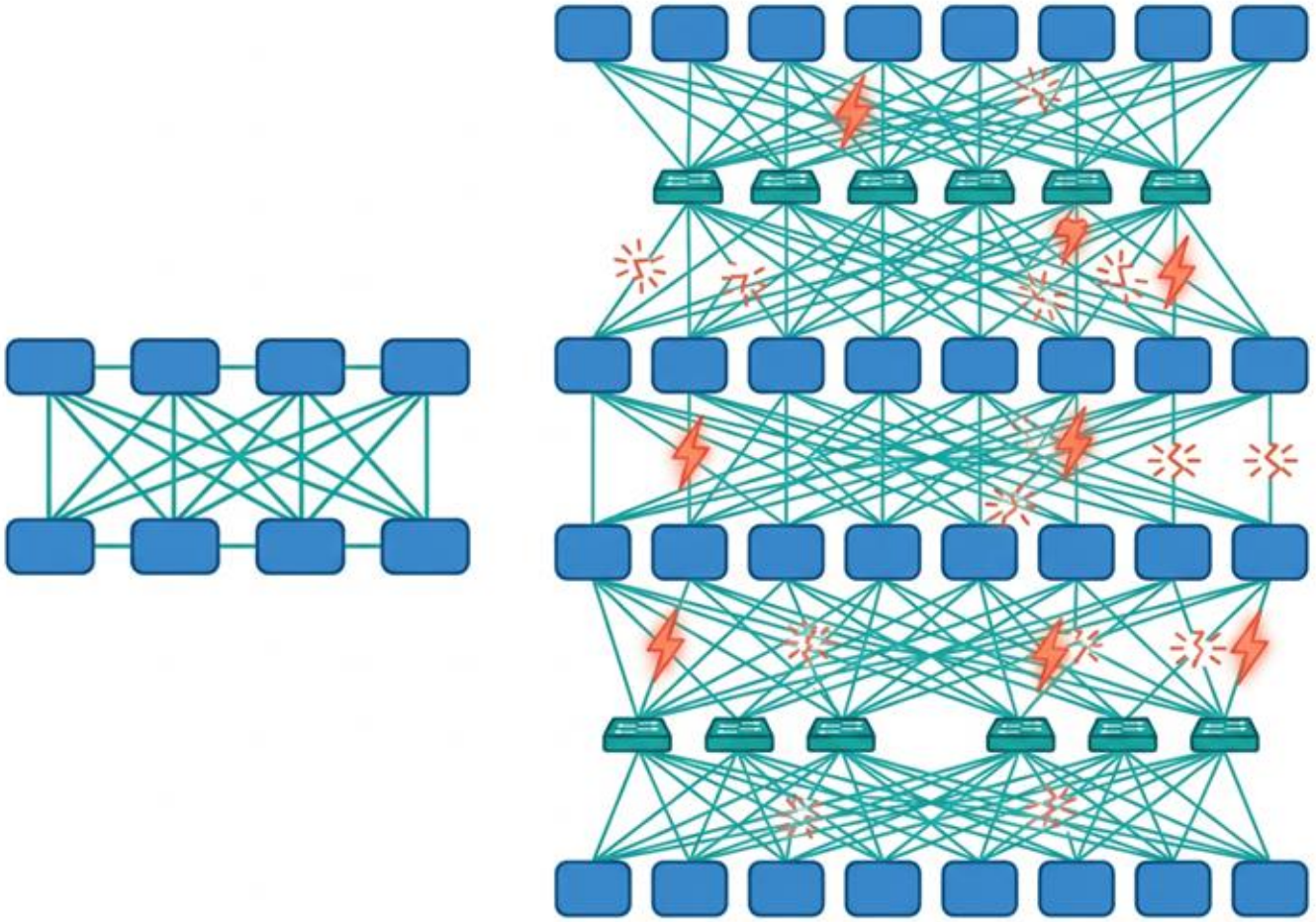
High Speed Cable Cartridge AI System Validation

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OCP PLATINUM Member

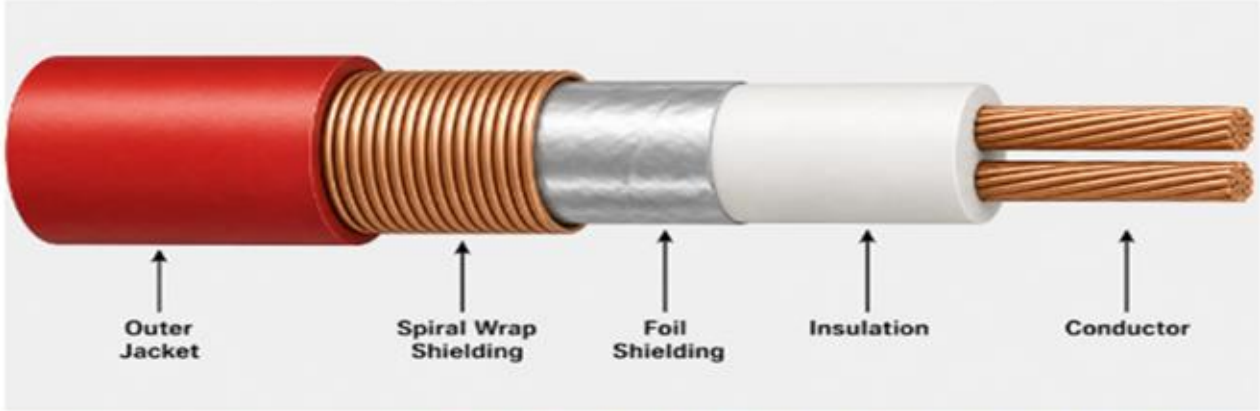
Importance of Scale-Up Networks



- P(fail) = 1−(1−p)^N
- Any single link failure kills the job if there’s no redundancy.
- As components scale with # of GPUs, Failure compounds.
- MTBF shrinks to hourly at 1000+ GPUs

Copper Cable Construction

- **Conductors:** Solid or stranded copper wires
- **Insulation:** Plastic coating around each wire (controls impedance)
- **Shield:** Metal foil or braid wrapping for noise protection
- **Jacket:** Outer protective cover



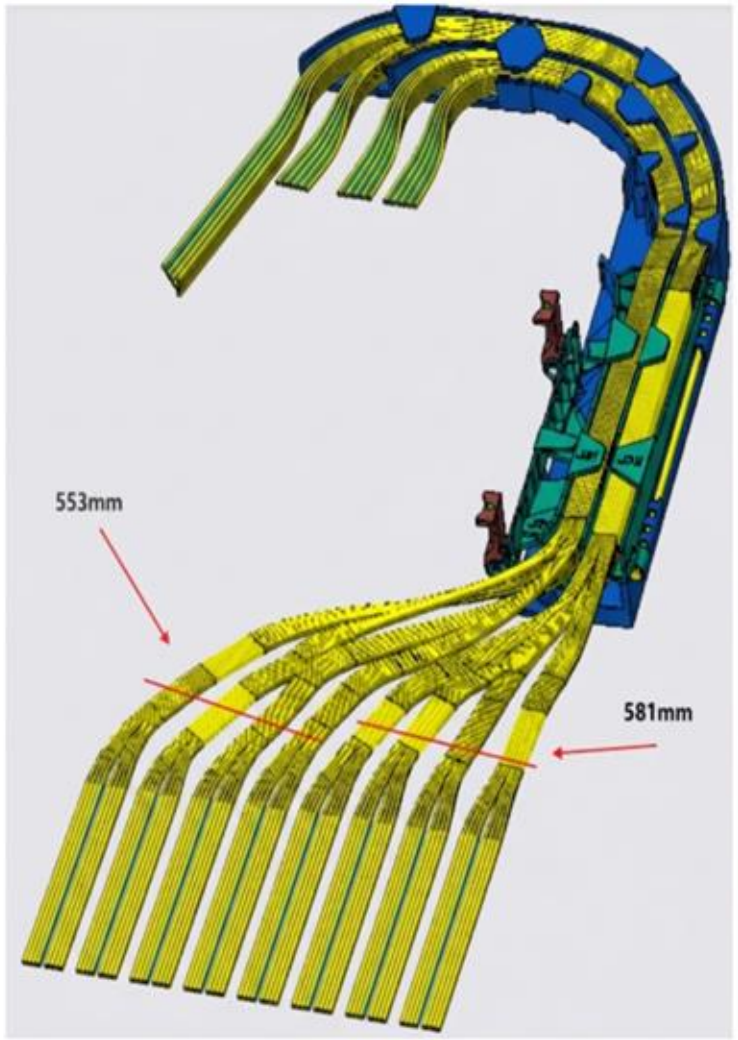
Advantages

- Low loss due to increased copper area and lower dielectric loss
- Loss + Length analysis can be done easily.

Key Challenges

- Bending changes electrical properties
- Connector interfaces add discontinuities
- Manufacturing variation between cables

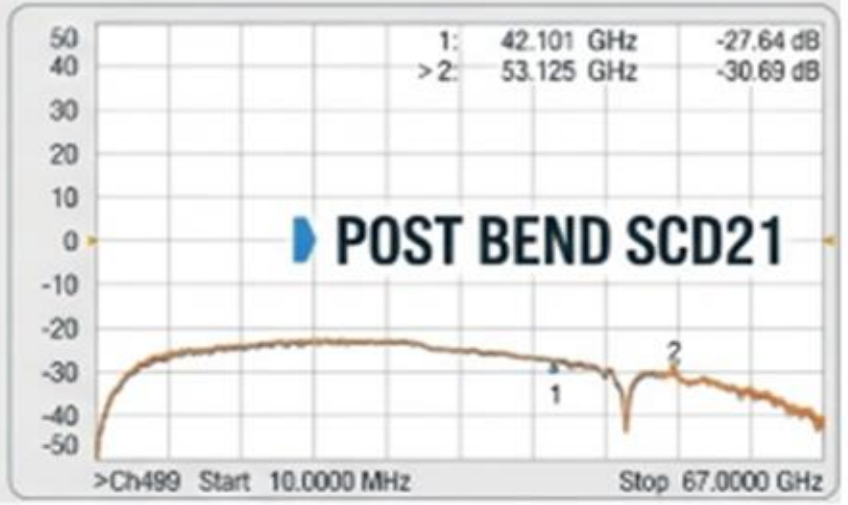
Cable Skew


553mm
581mm


1: 42,101 GHz -31.25 dB
>2: 53,125 GHz -33.40 dB
1: 42,101 GHz -33.49 dB
2: 33,125 GHz -36.91 dB
PRE BEND SKEW
Ch500 Start 10.0000 MHz
Stop 67.0000 GHz


>1: 46,168 GHz -30.34 dB
2: 53,123 GHz -33.73 dB
1: 46,168 GHz -48.37 dB
2: 53,125 GHz -39.24 dB
POST BEND SKEW
>Ch499 Start 10.0000 MHz
Stop 67.0000 GHz

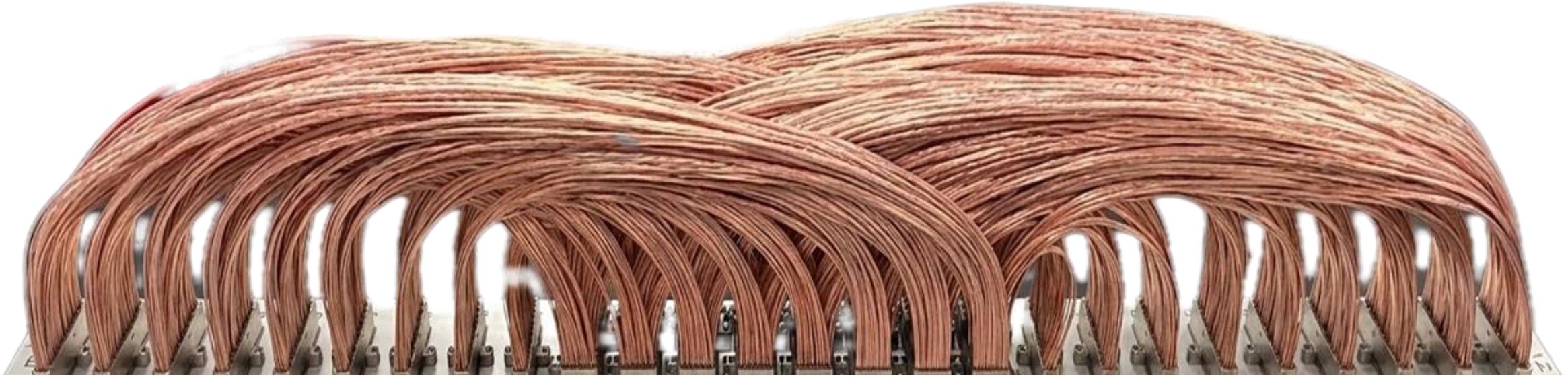

1: 42.101 GHz -41.70 dB
>2: 53.125 GHz -43.38 dB
PRE BEND SCD21
Ch500 Start 10.0000 MHz
Stop 67.0000 GHz


1: 42.101 GHz -27.64 dB
>2: 53.125 GHz -30.69 dB
POST BEND SCD21
>Ch499 Start 10.0000 MHz
Stop 67.0000 GHz

Cable Cartridge POC stage

- Reliable physical connections are essential for any-to-any connectivity between compute and switch trays.
- The design phase begins with testing boards and measuring S-parameters to validate performance.
- Key performance metrics measured include signal loss, skew, insertion loss deviation (ILD), and the impact of cable bending.
- Mapping from end-to-end is developed during the design phase to ensure maximum equalizable end-to-end loss.
- These measurements and mapping determine the required maximum cable length and bend radius.

Cable Cartridge during mockup



Bending Impact: Detailed Statistical Metrics

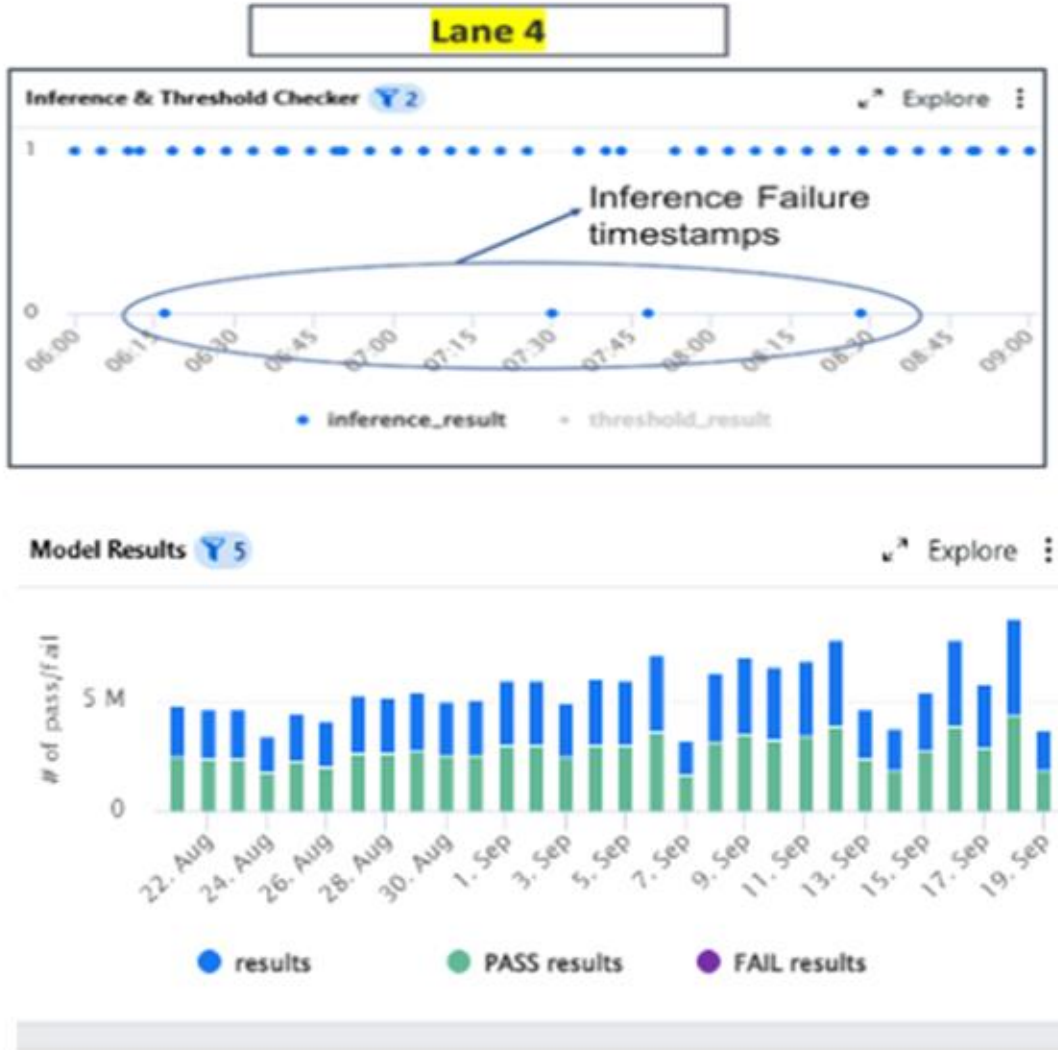
Metric	n	Mean Δ	% change	Effect (dz)	% degraded	Verdict
Eye Width (UI)	320	0.007	**+4.8%**	+0.17 (negligible)	44%	significant but tiny (negligible)
Eye Width (UI)	320	-0.001	-0.6%	-0.02 (negligible)	53%	no significant change
Eye Height (mV)	320	-0.061	-0.3%	-0.01 (negligible)	52%	no significant change
Eye Height (mV)	320	-0.063	-0.3%	-0.01 (negligible)	51%	no significant change
COM (dB)	320	0.341	**+6.2%**	+0.30 (small)	40%	**significant — bending improves (small)**
COM (dB)	320	0.047	+0.8%	+0.05 (negligible)	49%	no significant change
Eye W/H (UI/mV)	320	0.000	+4.4%	+0.13 (negligible)	44%	no significant change
Eye W/H (UI/mV)	320	-0.000	-0.3%	-0.01 (negligible)	49%	no significant change
EIPS |Skew| (ps)	320	1.642	**+325.6%**	+2.10 (large)	97%	**significant — degrades (large)**
EIPS |Skew| (ps)	320	0.369	**+63.9%**	+0.53 (medium)	68%	**significant — degrades (medium)**
SCMR 802.3dj (dB)	320	-10.069	**-44.0%**	-2.04 (large)	97%	**significant — degrades (large)**
SCMR 802.3dj (dB)	320	-2.443	**-11.1%**	-0.43 (small)	69%	**significant — degrades (small)**
IL @ Nyquist (dB)	320	-0.676	**+5.1%**	-1.21 (large)	89%	**significant — degrades (large)**
IL @ Nyquist (dB)	320	-0.018	+0.1%	-0.06 (negligible)	48%	no significant change

Cable Cartridge EVT, DVT stage

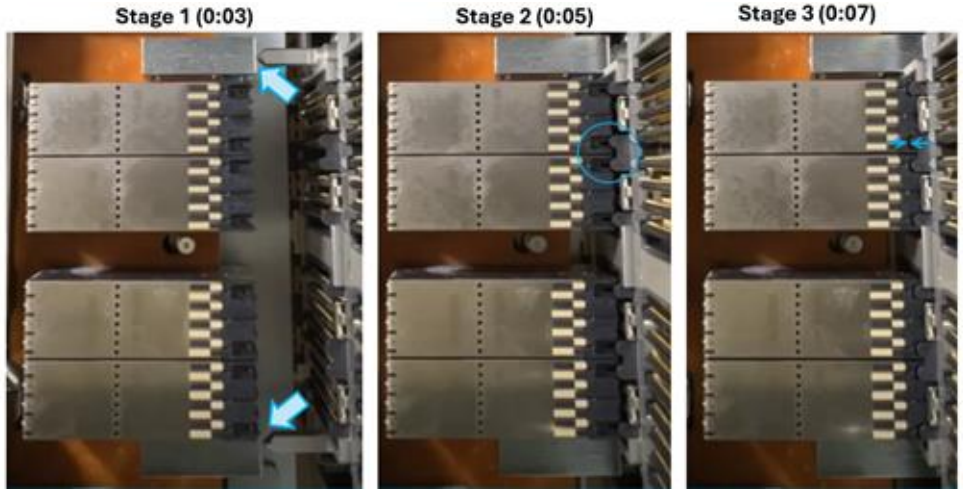
- Measure the BER floor, maximum FEC T, using idle patterns, traffic, and PRBS31Q at nominal voltage and temperature.
- Extract all relevant SerDes parameters for deeper analysis to pinpoint channels prone to burst errors or low margins.
- Meta’s Skew Analysis tool to determine acceptable skew from cables. This is used as a baseline for pre-bent cables acceptance criteria.
- Conduct 50 power cycle (on/off) experiments to verify SerDes stability.
- Replicate these measurements using the four corner test procedure (temperature and humidity) as described in next slide.

INSPECT - Interconnect System Parametric Checker Tool

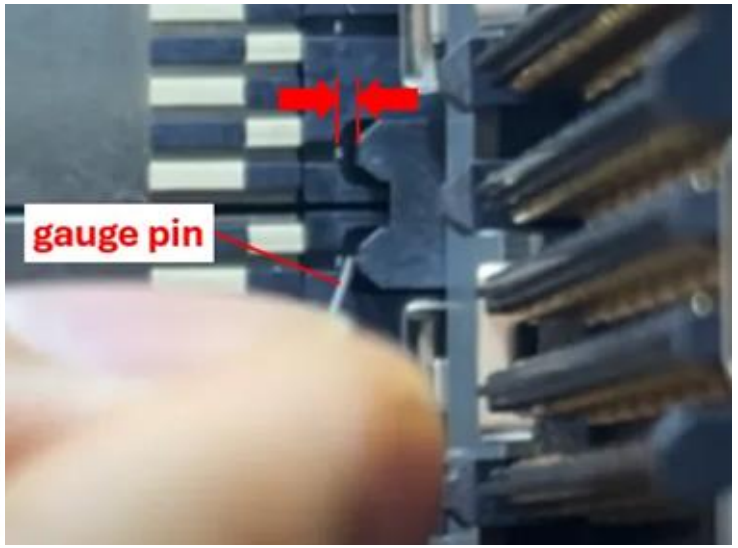
- INSPECT is a statistical analysis tool for SerDes channels that can be used for:
  - **Proactive maintenance** - detecting degrading links before complete failures
  - **Targeted repairs** - reducing blast radius by pinpointing fault locations
  - **Faster root cause analysis** - equalization parameters reveal specific channel impairments
  - **Serviceability improvements** - enabling component-level rather than system-level replacements



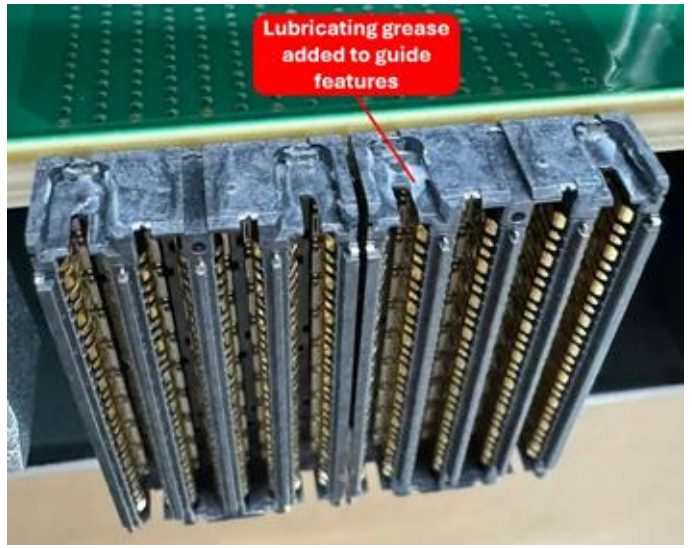
Mechanical Considerations



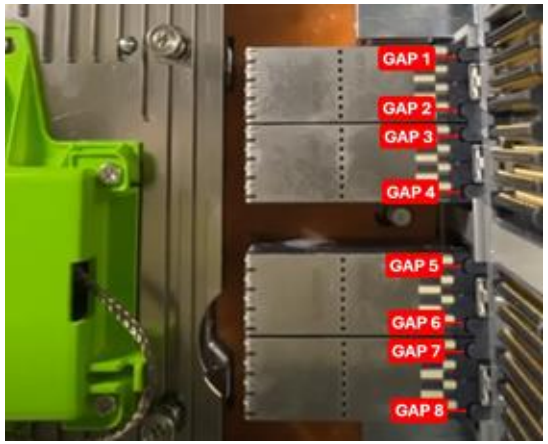
X/Y Connector Float Checks

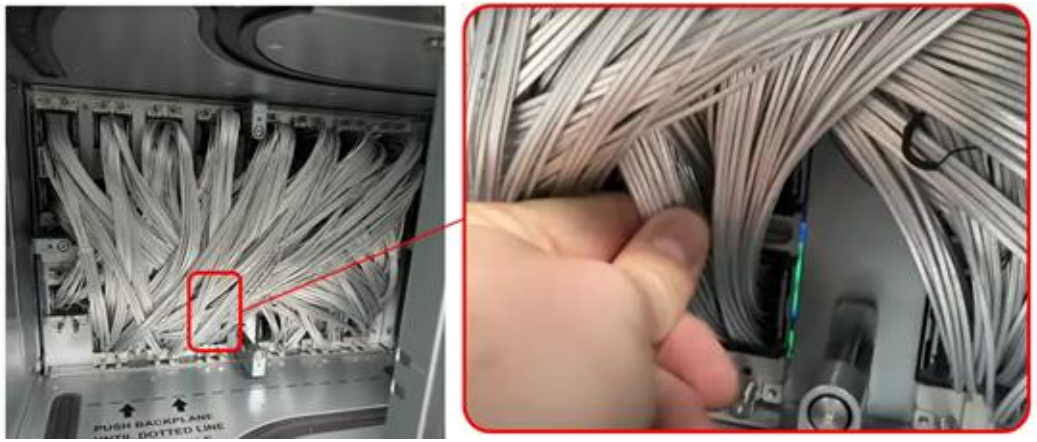


Connector Block Demate



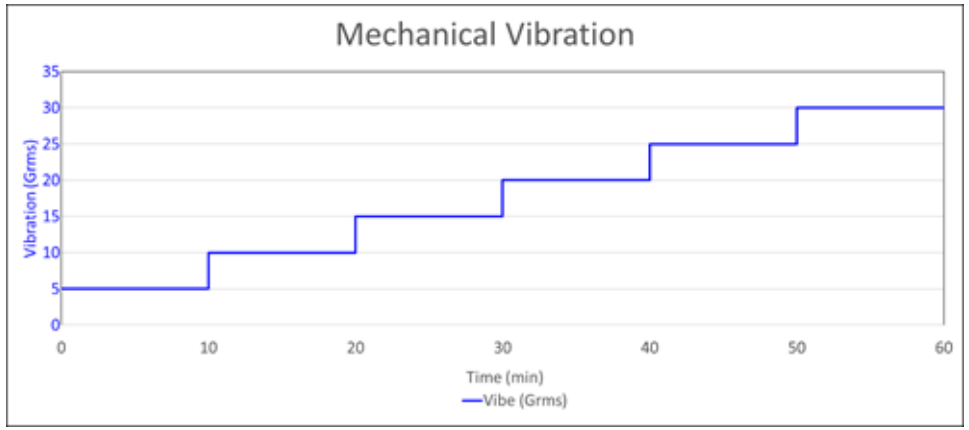
Lubricating Alignment urface





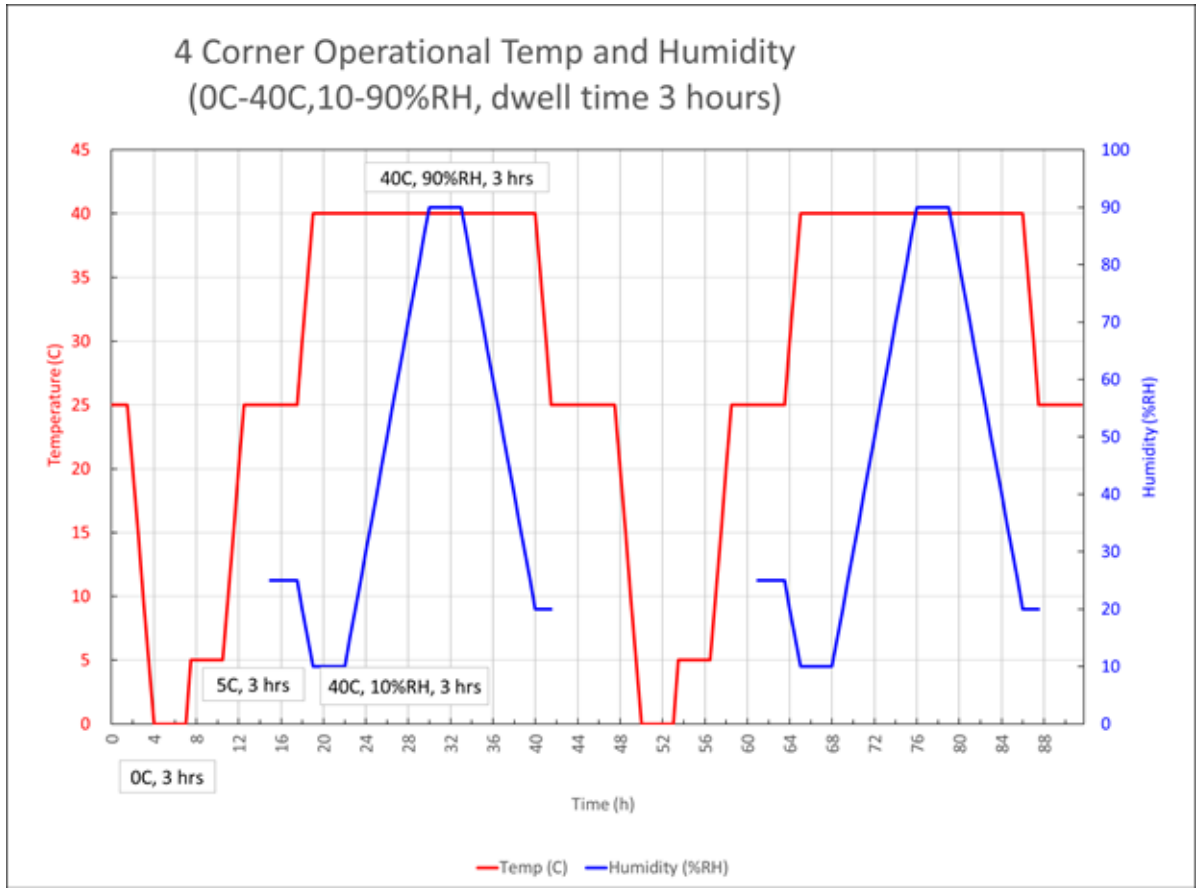
Durability, Stress Testing

- Durability, Stress Testing (S&V, Thermal, etc.)



Test Item	Test SPEC
OP Vibration	$0.5G_{rms}$, 5-500-5Hz, Random Vibe, 20 min along all three axes (+/-) 5-20 Hz – 6db/oct 20-250 Hz – 0.0007 G^2/Hz 250-500 Hz - -6db/oct
OP Shock	2G-6G, Half-Sine, 11ms, total 6 shocks, Along three axes (+/-)
Non-OP Vibration	1.2G, 5-500-5Hz per sweep, 1 sweep at 0.5 Octave/min, along all three axes (+/-) 5-10 Hz – 0.5G 10-350 Hz – 1.2G 350-500 Hz – 0.5G
Non-OP Shock	12G, Half-Sine, 11ms, total 6 shocks, Along all three axes (+/-)

Environmental Validation



- Run Operational temp & humidity
- Utilize different stepping of Silicon for worst and best case margins (SS, FF, TT)

Vendor Optimizations for Scale

- Vendor is ultimate quality gate for deployments at scale
- Cartridge vendor will run functional testing (BERT, TDR) and provide report against shipped cartridges
- Cartridge integrator will need to have earlier PRBS/BERT testing (EVT, DVT) and transition to functional testing (RDMA, etc.)
- RMA criteria needs to be aligned against both integrator and vendor!

Mass Production Tooling

- Repair actions are need to be clear and map to pod availability
- Post-repair screening is required (mechanical impulse = possible link degradation)
- RMA criteria to vendor needs to be aligned. PRBS, TDT, etc.

Call to Action

Alignment of our Validation Strategies for Cable Cartridges will expedite overall development and stability of cartridge for AI Systems Solutions across the industry

01
Simulation

Common connector choices, SerDes speeds, physical constraints

02
Physical Validation

Common test specifications against environmental, S&V,

03
Scaling

Aligning quality metrics against vendors and integrators, defined set of fleet tooling and health metrics

Strong partnerships with our Cartridge Vendors and Integrators will pay dividends in our platform NPIs

Thank You!

